

# Homework #5

## CS5402 — Intro To Data Mining

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Due Date: July 24<sup>th</sup>, 2018

### 1 DBSCAN

a) The table would look as follows:

| Point | x | y | Density | Designation  |
|-------|---|---|---------|--------------|
| A1    | 1 | 4 | 3       | Core Point   |
| A2    | 5 | 2 | 3       | Core Point   |
| A3    | 4 | 3 | 3       | Core Point   |
| A4    | 5 | 6 | 2       | Border Point |
| A5    | 2 | 5 | 2       | Border Point |
| A6    | 5 | 4 | 4       | Core Point   |
| A7    | 1 | 2 | 2       | Border Point |
| A8    | 3 | 1 | 1       | Noise        |

b) The points in the two clusters would be:

- {A1, A5, A7}
- {A2, A3, A4, A6}

The clusters are formed by first identifying the main points; this is done via calculating the  $\epsilon$ , or the density of the points (i.e., how many points are around those points). After the core points have been identified, the border points are found via observing which points are close (within an  $\epsilon$  value) to the core points. Finally, the remaining points are simply noise.

### 2 Bagging Vs Boosting

#### 2.1 Iris

Looking at the confusion matrix of using bagging:

```

a  b  c  <-- classified as
50  0  0 |  a = Iris-setosa
  0  0 50 |  b = Iris-versicolor
  0  0 50 |  c = Iris-virginica

```

And comparing to the confusion matrix of boosting:

```

a  b  c  <-- classified as
50  0  0 |  a = Iris-setosa
  0 45  5 |  b = Iris-versicolor
  0  1 49 |  c = Iris-virginica

```

We see that boosting is far more accurate; the kappa statistic agrees, where bagging had a kappa statistic of 0.5 versus the boosting kappa statistic of 0.94.

## 2.2 Iris

Looking at the confusion matrix of bagging:

```

445  13 |  a = benign
   7 234 |  b = malignant

```

And comparing to the confusion matrix of the boosting:

```

a  b  <-- classified as
445 13 |  a = benign
 17 224 |  b = malignant

```

We see there is very little difference in accuracy. The kappa statistic agrees; where bagging had a kappa statistic of 0.937 versus the boosting kappa statistic of 0.9046.

### 3 KMeans In R

assets/problem\_3.pdf

## **4 DBSCAN In R**



## **5 Frames For Days**

## **6 She's So Mean**

## **7 Animals**